

PROMPT TEMPLATES CHEATSHEET

AI Engineer Ready · Week 3 — Writing Reusable Prompts

Golden Rule → write the instruction once → fill

per call → always print before sending

1. Why Templates Instead of Raw Strings?

Hard-coding a prompt means rewriting it every time an input changes. A template lets you write the instruction **once** and plug in different values each call.

2. Basic Template with `.format()`

```
template = "Translate the text from {source_lang} to {target_lang}:\n\n{text}"

prompt = template.format(
    source_lang="English", target_lang="Spanish", text="LangChain is helpful."
)
```

RULE	DETAIL
Every <code>{name}</code> must be supplied	Missing one → <code>KeyError</code>
Literal curly brace	Double it: <code>{{</code> and <code>}}</code> (needed for JSON examples)
Multi-line prompts	Use triple-quoted strings <code>'''...'''</code>

3. Multi-Variable Template (Translation Example)

```
template = '''Translate the following text from {source_lang} to {target_lang}:

{text}

Provide only the translated text.'''

formatted = template.format(source_lang="English", target_lang="Spanish",
                           text="LangChain helps structure LLM apps.")
resp = chat.invoke(formatted)
print(resp.content)
```

4. Few-Shot Prompting Pattern

Show the model one worked example before asking for the real task — dramatically improves consistency, especially for structured output.

```
few_shot_template = '''You are a JSON responder. Return keys 'summary' and 'language'.

Example:
Input: I love programming in Python.
Output: {"summary": "likes programming", "language": "English"}

Now respond to:
Input: {text}
Output:'''
```

Pattern: instruction → example(s) → the real input → "Output:" as a nudge to stop extra commentary.

5. Common Mistakes

MISTAKE	SYMPTOM	FIX
Forgot a variable	<code>KeyError: 'text'</code>	Pass every <code>{name}</code> used
Single <code>{</code> in JSON example	<code>KeyError</code> / garbled output	Escape as <code>{{ }}</code>
Didn't check formatted output	Model output subtly wrong	Print the formatted prompt before sending
Vague instructions	Inconsistent format	Be explicit + add a few-shot example

Checklist Before Sending a Prompt

- ☐ Printed the formatted prompt and it reads as intended
- ☐ Instructions are specific ("only the translated text", not just "translate")
- ☐ Structured output requests include a worked example